

BTC Oracle

ML System & Infrastructure

Technical Analysis Report

Training Pipeline | Data Sources | Cost Analysis | Architecture Deep-Dive

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1. Executive Summary

BTC Oracle is a production-grade Bitcoin price prediction system that combines four machine learning models in a weighted ensemble, augmented by a 15-signal quantitative engine. The system ingests data from 20+ collectors across 10+ free APIs, processes 222 engineered features per hour, and generates predictions across five time horizons (1h, 4h, 24h, 1w, 1mo).

The platform includes continuous self-learning (every 6 hours), A/B model testing, pattern discovery, and adaptive ensemble weight adjustment. It is deployable via Docker on minimal hardware and can operate entirely on free APIs at near-zero cost, or be enhanced with premium data sources for improved accuracy.

4 ML MODELS	222 FEATURES	20+ COLLECTORS	5 TIMEFRAMES
\$10-50 MONTHLY COST (Free APIs)	~500 MB YEARLY STORAGE	2-3s INFERENCE TIME	6h LEARNING CYCLE

2. System Architecture Overview

The system follows a modular pipeline architecture with clear separation between data collection, feature engineering, model inference, and evaluation.

Data Pipeline Flow

- 1. Raw Data Sources (Binance, blockchain.info, RSS feeds, FRED, CoinGecko, etc.)
- 2. 20 Async Collectors (aiohttp-based, scheduled via APScheduler)
- 3. Feature Engineering (222 features computed per hourly snapshot)
- 4. ML Ensemble (TFT + LSTM + XGBoost + TimesFM, weighted voting)
- 5. Quant Engine (15+ theory-based signals, composite scoring)
- 6. Predictions & Signals (stored in DB, served via FastAPI REST API)
- 7. Evaluation & Learning (accuracy tracking, pattern discovery, weight adaptation)

Technology Stack

Component	Technology	Version
Backend Framework	FastAPI (async Python)	0.115.6
Database ORM	SQLAlchemy (async)	2.0.36
Database	SQLite (dev) / PostgreSQL (prod)	3.x / 16+
ML Framework	PyTorch + PyTorch Lightning	2.x
Tree Ensemble	XGBoost	2.1.3
Foundation Model	Google TimesFM 2.0	200M params
Technical Analysis	ta library + custom	0.11.0
Scheduling	APScheduler	3.10.4
Frontend	React + Vite + Tailwind CSS	18.x
Deployment	Docker Compose	Multi-service

3. ML Models & Ensemble

The core prediction engine uses a weighted ensemble of four diverse model architectures. Each model processes the same 222-feature input but through different mechanisms, providing robust predictions through model diversity.

Model	Architecture	Input Shape	Parameters	Training Time
TFT	Temporal Fusion Transformer 4 attention heads, 64 hidden dim, 3 LSTM layers, multi-horizon output	(168, 222) 7-day sequence	~500K-1M	5-30 min (GPU preferred)
LSTM	Bidirectional LSTM 3 layers, 128 hidden, 0.2 dropout, multi-head output	(168, 222) 7-day sequence	~300K-500K	5-20 min
XGBoost	Gradient Boosted Trees max_depth=6-8, 100-200 estimators, binary:sigmoid objective	(1, 222) current snapshot	N/A (trees)	1-5 min (CPU only)
TimesFM	Google Foundation Model pre-trained 200M params, zero-shot forecasting	Price history (variable length)	200M (frozen)	No training (inference only)

Ensemble Voting Mechanism

Predictions from all four models are combined using adaptive weights that update every 6 hours based on recent accuracy. The weight formula is: **weight = accuracy^2** (normalized to sum to 1.0), which exponentially rewards better-performing models. When a model's accuracy drops below 50%, it triggers selective retraining on the last 72 hours of data.

Prediction Outputs (Per Timeframe)

Timeframe	Schedule (UTC)	Output Fields
1 hour	Every hour at :00	Direction (bullish/bearish), confidence, predicted price, predicted change %
4 hours	00:00, 04:00, 08:00, 12:00, 16:00, 20:00	Same fields, medium-term outlook
24 hours	Daily at 00:04	Same fields, daily trend prediction
1 week	Via quant engine	Composite score, target price range
1 month	Via quant engine	Long-term directional bias, cycle position

4. Feature Engineering (222 Features)

Every hour, the system computes 222 features from raw market data. Features are organized into 15 categories spanning technical analysis, sentiment, macro economics, on-chain metrics, derivatives, and more. All features are z-score normalized using saved training parameters.

Category	Count	Key Examples
Technical Indicators	111	EMA(9,21,50,200), RSI, MACD, Bollinger Bands, ATR, Ichimoku, Stoch RSI, ADX, GARCH, Hurst, Wavelets
Sentiment	13	News sentiment (1h/4h/24h), Reddit, Fear & Greed, social bullish/bearish %
Derivatives Extended	12	Long/short ratio, DVOL, liquidations (long/short), estimated leverage, taker buy/sell
Whale Activity	12	TX count (1h/24h), exchange in/out, net flow, severity, directional signal
Power Law & Long-Term	11	Deviation from fair value, 200d SMA ratio, 52w high/low, halving cycle position
Exchange Flows	9	Reserve BTC, netflow, NVT, MVRV Z-score, SOPR, Puell, supply in profit, LTH supply
Macro Economics	8	DXY change, Gold change, S&P500 change, Treasury 10Y yield, M2 supply
On-Chain	7	Hash rate, mempool size/fees, TX volume, active addresses, difficulty
Event Memory	7	Expected impact (1h/4h/24h), confidence, severity, active event count
ETF Flows	6	Net flow USD, total holdings BTC, IBIT/FBTC/GBTC individual flows
Stablecoin Supply	5	USDT/USDC market cap, total supply, 7d change %, DeFi TVL
Market Dominance	4	BTC dominance, ETH dominance, total market cap, 24h change
Derivatives Basic	3	Funding rate, open interest, mark-index spread
Phrase Correlation	3	Top bullish/bearish phrase scores, net signal
Supply / Mining	4	% mined, daily issuance, blocks until halving, halving cycle %

Sequence Construction for Neural Models

LSTM and TFT models require sequential input. The system builds sliding windows of **168 hourly snapshots (7 days)**, each containing all 222 features. This produces input tensors of shape **(batch, 168, 222)**. Sequences shorter than 168 are left-padded with zeros. XGBoost uses only the current snapshot (1, 222).

5. Data Sources & Collectors

The system uses 20 asynchronous collectors to gather data from diverse sources. All primary data sources are free, with optional premium sources for enhanced accuracy.

Collector	Source(s)	Interval	Cost	Key Data
Market	Binance REST API	60s	Free	OHLCV, funding rate, OI, premium index
On-Chain	blockchain.com, mempool.space	1 hour	Free	Hash rate, mempool, TX volume, addresses
News	35+ RSS feeds	2 min	Free	Headlines from CoinDesk, Cointelegraph, etc.
Fear & Greed	alternative.me	1 hour	Free	Daily F&G index (0-100)
Macro	FRED, CoinGecko	1 hour	Free	DXY, Gold, S&P500, Treasury yields, M2
Reddit	PRAW API	1 hour	Free	Post titles, upvotes, sentiment
Influencers	Social media feeds	10 min	Free	Posts from known BTC influencers
ETF Flows	CoinGlass, SoSoValue	1 hour	Free	IBIT/FBTC/GBTC flows, total holdings
Derivatives	Binance Futures, Deribit	1 hour	Free	Liquidations, leverage, DVOL
Stablecoin	CoinGecko, DeFi Llama	1 hour	Free	USDT/USDC supply, DeFi TVL
Whale	mempool.space, BTCScan	10 min	Free	Large TX tracking (>100 BTC)
Exchange Flows	Glassnode, blockchain.com	1 hour	Optional	NVT, MVRV, SOPR, reserves
Coins	CoinGecko	2 min	Free	Altcoin prices, market cap, dominance
Historical	Binance, yfinance	On demand	Free	Full OHLCV history since 2009

Free APIs Currently Used (No Keys Required)

- Binance REST API (public market data, unlimited)
- CoinGecko API (10-50 calls/min, sufficient for hourly updates)
- blockchain.info (free stats and on-chain data)
- mempool.space (free mempool and block data)
- alternative.me (Fear & Greed Index, free)
- FRED (Federal Reserve data, requires free API key)
- DeFi Llama (TVL data, free)
- CoinGlass (ETF and liquidation data, public endpoints)

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- Deribit (DVOL index, public endpoints)
 - 35+ RSS feeds (crypto news sources, unlimited)

6. Training Pipeline

How Training Works (Step by Step)

Step 1: Dataset Construction — Pull all Feature snapshots (hourly) and corresponding Price records from the database. Build sliding window sequences of 168 hours (7 days).

Step 2: Label Generation — For each sample at timestamp T, compute the actual price at T+1h, T+4h, T+24h, T+1w, T+1mo. Generate binary direction labels (bullish/bearish) and magnitude percentages for each horizon.

Step 3: Data Split — 80% training, 20% validation (chronological split to prevent data leakage from future information).

Step 4: Model Training — TFT and LSTM train on sequences (168, 222) with multi-task loss across all 5 timeframes. XGBoost trains on current features (1, 222) with binary classification. TimesFM requires no training (pre-trained foundation model).

Step 5: Validation & Storage — Evaluate on holdout set, save weights to persistent volume (/data/weights/), record metrics in ModelVersion table.

Step 6: A/B Testing — New model registered as candidate. Must outperform production model by 3% over 24+ hours and 50+ predictions to be promoted.

Training Resource Requirements

Component	GPU Memory	CPU Time	Frequency	Notes
TFT Training	4-6 GB	5-30 min	24-48h	GPU preferred but CPU works
LSTM Training	2-4 GB	5-20 min	24h	Bidirectional, 3 layers
XGBoost Training	<1 GB	1-5 min	24h	CPU only, very fast
TimesFM Inference	4 GB	5-10s (GPU) 20-30s (CPU)	Per prediction	Pre-trained, no training needed
Feature Extraction	<1 GB	0.5-2s	Every hour	222 features from raw data
Continuous Learning	<1 GB	1-5 min	Every 6h	Weight adaptation + selective retrain

7. Continuous Learning & A/B Testing

The system implements three layers of self-improvement that run automatically:

7.1 Adaptive Ensemble Weights (Every 6 Hours)

Computes rolling accuracy for each model over the last 48 hours. Weights are set to **accuracy²** (squared to exponentially reward top performers) and normalized to sum to 1.0. Minimum 20 predictions per model required.

7.2 Selective Retraining

If any model's accuracy drops below 50%, only that model is retrained on the last 72 hours of data. The new weights are validated against a holdout set. If improved, they are hot-swapped; if not, the current weights are retained. This prevents catastrophic degradation while allowing recovery.

7.3 Pattern Discovery (Every 6 Hours)

Five pattern types are automatically discovered and applied:

Pattern Type	What It Detects	How It's Applied
Volatility Regime	Accuracy varies by volatility level	Confidence modifier when high/low volatility detected
Confidence Calibration	Stated confidence vs actual accuracy	Recalibrate confidence output to match reality
Model Disagreement	Accuracy drops when models disagree	Reduce confidence when agreement is low
Feature Thresholds	Specific feature combos predict well	Boost/reduce confidence when pattern active
Time Patterns	Accuracy varies by time of day	Adjust predictions during known weak/strong periods

7.4 A/B Testing Protocol

Parameter	Value
Minimum test duration	24 hours
Minimum predictions	50
Promotion threshold	3% accuracy improvement over production
Maximum test duration	72 hours (auto-reject if no improvement)
Rollback	Automatic if candidate underperforms

8. Quantitative Prediction Engine

In addition to ML models, a separate quantitative engine generates predictions using 15+ theory-based signals from established Bitcoin valuation frameworks. This provides an independent second opinion that is not dependent on training data.

Signal	Weight	Description
Pi Cycle Top Indicator	8%	111DMA vs 350DMA×2 crossing (historical top detector)
Mayer Multiple	8%	Price / 200DMA ratio (<0.6 = buy, >2.4 = sell)
Halving Cycle Position	8%	Position within 4-year halving cycle (0-1460 days)
Mean Reversion Z-Score	8%	Statistical reversion at ±2.5 standard deviations
Momentum (20d vs 100d)	8%	Short vs long-term return comparison
Rainbow Chart Bands	7%	9 valuation bands from fire sale to max bubble
DX/RSI Inverse Correlation	7%	USD strength inverse relationship (-0.72 correlation)
Fear & Greed Contrarian	7%	Buy at extreme fear, sell at extreme greed
Funding Rate Extremes	7%	Extreme rates precede 70%+ of corrections
Regime Detection	7%	Volatility ratio + SMA50 market regime
NVT Ratio	5%	Network value vs transaction value (<30 = undervalued)
Puell Multiple	5%	Miner revenue vs 365d average
Volume Profile (POC)	5%	Point of Control, Value Area analysis
Round Number S/R	5%	Psychological support/resistance levels
ATH Proximity	5%	Distance from all-time high (momentum gauge)

The composite score ranges from **-100** (extremely bearish) to **+100** (extremely bullish). Action thresholds: >60 = STRONG_BUY, >30 = BUY, ±10 = NEUTRAL, <-30 = SELL, <-60 = STRONG_SELL.

9. Infrastructure & Deployment

Current Deployment Architecture

Component	Specification	Details
Runtime	Docker Compose	2 services: backend + webapp
Backend Image	Python 3.12-slim	FastAPI, PyTorch, XGBoost, SQLAlchemy
Frontend	React + Vite	Tailwind CSS, i18n (EN/RU/ZH)
Database (Dev)	SQLite +aiosqlite	Single file, zero config
Database (Prod)	PostgreSQL + asyncpg	Connection pooling, async sessions
Model Storage	/data/weights/	Persistent Docker volume
Backup System	6-hour intervals	7-day retention, SQLite snapshots
Scheduling	APScheduler	40+ jobs, cron + interval triggers
Bot Integration	Telegram (aiogram)	Polling mode, webhook support

Minimum Server Requirements

Tier	CPU	RAM	Storage	GPU	Monthly Cost	Best For
Minimal	1 vCPU	1 GB	5 GB	None	\$5-10	Development, testing
Standard	2 vCPU	4 GB	20 GB	None	\$20-40	Production (CPU inference)
Enhanced	4 vCPU	8 GB	50 GB	T4 (16GB)	\$50-150	Fast training + TimesFM
Premium	8 vCPU	16 GB	100 GB	A10G (24GB)	\$150-400	Full suite + custom models

Note: The system currently runs on the Standard tier without GPU. TFT and LSTM training takes 5-30 minutes on CPU. Adding a GPU reduces training to 1-5 minutes and enables fast TimesFM inference (5-10s vs 20-30s on CPU).

10. Cost Analysis & Projections

Current Monthly Operating Costs

Category	Item	Monthly Cost	Notes
Hosting	Railway / Render / AWS	\$10-50	1-2 vCPU, 2-4 GB RAM
Database	PostgreSQL (managed)	\$10-25	Optional upgrade from SQLite
API: Binance	Market data	\$0	Free, no key required
API: CoinGecko	Prices, market cap	\$0	Free tier sufficient
API: blockchain.info	On-chain data	\$0	Free
API: mempool.space	Mempool, blocks	\$0	Free, open-source
API: FRED	Macro economics	\$0	Free API key
API: CoinGlass	ETF flows, liquidations	\$0	Public endpoints
API: RSS Feeds	35+ news sources	\$0	Standard RSS
API: alternative.me	Fear & Greed	\$0	Free

Total (Free APIs only): \$10 - \$50/month

Optional Premium Additions

Service	What It Adds	Monthly Cost	Impact on Accuracy
Glassnode Professional	SOPR, MVRV Z-score, NVT signal, exchange reserves, entity flows	\$500-1,000	High: critical on-chain metrics not available elsewhere
Nansen Basic	Address identification, smart money tracking, entity labels	\$0-99	Medium: improved whale classification
GPU Instance (T4)	Faster training (5x), TimesFM inference, future model expansion	\$40-100	Medium: enables TimesFM and faster iterations
Dedicated PostgreSQL	Better concurrency, ACID compliance, scalability beyond 1 GB	\$25-50	Low: reliability improvement, not accuracy

Total (with Glassnode + GPU): \$550 - \$1,200/month

11. Storage & Growth Forecast

Component	Current Size	Annual Growth	3-Year Projection
Price Data (hourly OHLCV)	~50 MB	~9 MB/year	~77 MB
Feature Snapshots (222 features, JSON)	~300 MB	~500 MB/year	~1.8 GB
Predictions (all timeframes)	~20 MB	~2 MB/year	~26 MB
News & Sentiment	~30 MB	~50 MB/year	~180 MB
Whale Transactions	~10 MB	~20 MB/year	~70 MB
Model Weights	~100 MB	~50 MB/year	~250 MB
Backups (7-day rolling)	~3.5 GB	Constant	~3.5 GB
Logs	~50 MB	~100 MB/year	~350 MB

~500 MB

YEAR 1 DATABASE

~1.5 GB

YEAR 2 DATABASE

~2.5 GB

YEAR 3 DATABASE

~6 GB

TOTAL (WITH BACKUPS)

Storage growth is very manageable. Even at 3 years of operation, the total footprint (including rolling backups) stays under 10 GB. A 20 GB disk allocation is sufficient for 5+ years of continuous operation.

12. Recommendations for Custom AI

If the goal is to build a fully custom AI model that you own and can license, here is a realistic roadmap with cost estimates:

Phase 1: Data Foundation (Month 1-2)

Continue running the current system to accumulate high-quality training data. The more historical predictions with outcomes you have, the better a custom model will perform. Target: 3-6 months of hourly data with evaluation results.

Phase 2: Custom Model Development (Month 3-6)

Investment	Cost	What You Get
ML Engineer (contract)	\$5,000-15,000	Custom transformer architecture trained on your data
GPU Compute (training)	\$200-500	Cloud GPU hours (A100/H100) for training runs
Glassnode API	\$3,000-6,000	6 months of premium on-chain data for training
Data labeling / validation	\$500-1,000	Manual verification of training labels

Phase 3: Production Custom Model (Month 6+)

Scenario	Setup Cost	Monthly Cost	What You Own
DIY (current system + improvements)	\$0	\$50-100	Everything: code, weights, data pipeline
Custom transformer model	\$5,000-15,000	\$100-300	Proprietary model + training pipeline
Full custom AI platform	\$20,000-50,000	\$300-1,000	End-to-end system, licensable IP

Key Advantages of the Current System

- ✓ **You already own everything.** All code, models, and data pipelines are yours. No vendor lock-in.
- ✓ **Near-zero operating cost.** The system runs on \$10-50/month using only free APIs.
- ✓ **Self-improving.** Continuous learning, A/B testing, and pattern discovery run automatically.
- ✓ **Production-ready.** Docker deployment, Telegram bot, multi-language frontend, admin dashboard.
- ✓ **Extensible.** Adding new data sources or models is straightforward (follow existing patterns).

Bottom Line: The current BTC Oracle system is a fully functional, self-learning ML platform that you own outright. It costs \$10-50/month to operate. Building a custom AI on top of this foundation (rather than from scratch) is the most cost-effective path. Budget \$5K-15K for a custom model if you want to go beyond the current 4-model ensemble, or continue improving the existing system for free.